

Advanced (Habanero)

- Q1.** Evaluate $f(g(x))$ if $f(x) = x^2$ and $g(x) = 2x + 1$.
(A) $4x^2 + 4x + 1$
(B) $4x^2 + 1$
(C) $2x^2 + 2x + 1$
(D) $x^2 + 2x + 1$
(E) $2x^2 - 2x - 1$
- Q2.** Find the domain of $f(g(x))$ where $f(x) = \frac{1}{x}$ and $g(x) = x - 2$.
(A) $(-\infty, 2) \cup (2, \infty)$
(B) $(-\infty, 2)$
(C) $(-\infty, 0) \cup (0, \infty)$
(D) $[2, \infty)$
(E) $(-\infty, 0)$
- Q3.** What is $f(g(h(x)))$ if $f(x) = x^3$, $g(x) = 2x$, and $h(x) = x + 1$?
(A) $8x^3 + 12x^2 + 6x + 1$
(B) $8x^3 + 12x^2 + 6x$
(C) $8x^3 + 12x^2$
(D) $8x^3$
(E) $2x^3 + 6x^2 + 6x + 1$
- Q4.** Determine the value of x for which $f(g(x))$ is undefined, given $f(x) = \frac{x}{x-1}$ and $g(x) = x - 3$.
(A) $x = 4$
(B) $x = 2$
(C) $x = 3$
(D) $x = 1$
(E) $x = 0$
- Q5.** Find $f(g(0))$ if $f(x) = x^2 + 1$ and $g(x) = -x$.
(A) 2
(B) 1
(C) 0
(D) -1
(E) -2
- Q6.** Calculate $g(f(x))$ where $g(x) = 2x + 3$ and $f(x) = x^2$.
(A) $2x^2 + 3$
(B) $2x^2 - 3$
(C) $x^2 + 3$
(D) $x^2 - 3$
(E) x^2
- Q7.** If $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?
(A) $[-2, 2]$
(B) $[-2, 2) \cup (2, \infty)$
(C) $[2, \infty)$
(D) $[-\infty, -2) \cup [2, \infty)$
(E) $[-2, 2) \cup [2, \infty)$
- Q8.** For $f(x) = \frac{1}{x}$ and $g(x) = x + 2$, determine the domain of $f(g(x))$.
(A) $(-\infty, -2) \cup (-2, \infty)$
(B) $[-2, \infty)$
(C) $(-\infty, -2)$
(D) $(-2, \infty)$
(E) $[-2, 0)$

Open Ended Questions (FRQ)

- Q9.** Given $f(x) = x^2$ and $g(x) = \sqrt{x}$, evaluate $f(g(x))$ and $g(f(x))$. Compare the results.

- Q10.** Consider a water tank with an initial water level of 10 meters. Water is being drained at a rate of $f(x) = -2x$ m/min and filled at a rate of $g(x) = x^2$ m/min. Determine the net rate of change in the water level $h(x)$ after 5 minutes, given that $h(x) = f(g(x))$. Show all work and justify your answer.

Chef's Tips

- Always check the units of measurement in real-world applications.
- Ensure students understand the concept of function composition before moving on to domain restrictions.
- Remind students that in function composition, the order of functions matters.